

Magnetotorquer Design: The Math, the Logic, and the Numbers

Tulika Chaudhury

July 27, 2026

Why we need these at all

The satellite has three reaction wheels, each tilted at $\cos^{-1}(1/\sqrt{3}) \approx 54.74^\circ$ from the Z axis and spaced 120° apart in azimuth. Dotting the three spin-axis unit vectors pairwise gives zero in all three cases, confirming they're genuinely mutually orthogonal despite none of them lying along a body axis – a deliberate choice that gives equal torque and momentum capability in every direction using only three actuators.

Reaction wheels are excellent for fine pointing, but they only *move* angular momentum around – from the spacecraft body into the spinning wheel and back – they never get rid of it. Any small, steady external torque (gravity gradient, atmospheric drag, sunlight pressure, the satellite's own stray magnetism) makes the wheels absorb momentum faster than they release it, and their speed creeps up orbit after orbit until one *saturates* and stops being able to produce torque at all.

Magnetotorquers fix this because they push against something external – Earth's magnetic field – so they can actually remove momentum rather than just shuffle it. They do two jobs: detumbling right after deployment, when body rates are too high for the wheels to safely absorb, and periodic momentum “dumping” during normal operation to keep the wheels in their healthy range.

The physics, in plain terms

A current-carrying coil makes a magnetic dipole – a tiny switchable bar magnet – of strength

$$m = NIA, \tag{1}$$

where N is the turn count, I the current, and A the enclosed area. A magnetically soft core inside the coil concentrates the field and boosts this for the same current:

$$m = \mu_{\text{rod}} NIA_{\text{core}}. \tag{2}$$

Immersed in Earth's field $\mathbf{B}$, that dipole feels a torque

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}. \tag{3}$$

The cross product matters: torque is always perpendicular to $\mathbf{B}$, so a torquer can never twist the spacecraft about the direction the field happens to point at that instant. That's why three axes are needed – between them, and as $\mathbf{B}$ rotates through the body frame over an orbit, they cover everything.

Why a rod doesn't get the full benefit of its core

A finite rod partially cancels its own field at the ends – the same reason a short bar magnet is weaker than a long one of the same material. This self-demagnetization is captured by a demagnetizing factor N_d , which for a rod of length L , diameter d (aspect ratio $\ell = L/d$) is approximately

$$N_d \approx \frac{\ln(2\ell) - 1}{\ell^2}, \quad (4)$$

knocking the bulk material permeability μ_r down to an effective

$$\mu_{\text{rod}} = \frac{\mu_r}{1 + N_d(\mu_r - 1)}. \quad (5)$$

The shorter the rod, the harder this bites.

How the satellite decides when and how hard to fire the coils

Detumbling. With only a magnetometer to work from, the standard approach is the *B-dot law*:

$$\mathbf{m}_{\text{cmd}} = -k \dot{\mathbf{B}}_{\text{body}} \approx k (\boldsymbol{\omega} \times \mathbf{B}). \quad (6)$$

While tumbling fast, the field as seen by the satellite changes mostly because the satellite is spinning, not because the field itself moves. Commanding a dipole proportional to that apparent rate of change produces a torque that always opposes the part of the spin perpendicular to $\mathbf{B}$ – a magnetic brake needing nothing more than a magnetometer.

Momentum dumping. Once wheel momentum $\mathbf{h}_w$ needs shedding, the desired de-saturation torque is $\boldsymbol{\tau}_{\text{des}} = -k_h \mathbf{h}_w$. Since only the component perpendicular to $\mathbf{B}$ is achievable at any instant, the best available command is the projection

$$\mathbf{m}_{\text{cmd}} = \frac{\mathbf{B} \times \boldsymbol{\tau}_{\text{des}}}{|\mathbf{B}|^2}. \quad (7)$$

As $\mathbf{B}$ rotates through the body frame once per orbit, every component of $\mathbf{h}_w$ eventually gets its turn to be perpendicular to the field, so it all comes off over time.

How big does the dipole need to be?

Table 1: Worst-case steady disturbance torque, 500 km LEO.

Source	Size	Notes
Gravity gradient	20.6 nN m	from the CAD inertia tensor
Aerodynamic drag	43.2 nN m	2 cm CG/CP offset
Solar radiation pressure	4.9 nN m	smallest contributor
Residual spacecraft dipole	400 nN m	dominates ; 0.01 A m ² residual
Total	469 nN m	linear (conservative) sum

The satellite's own stray magnetism – current loops in the avionics stack – dominates this budget, outweighing gravity, drag, and sunlight pressure combined (Figure 1).

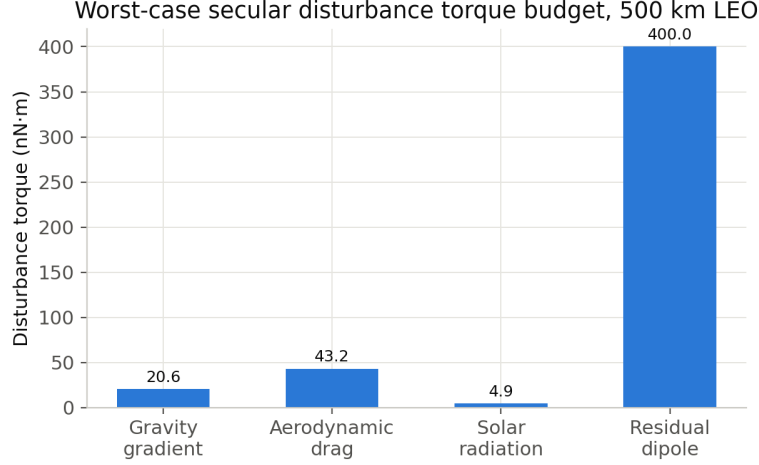


Figure 1: Where the disturbance torque comes from.

Sizing rule: double the disturbance torque for margin, and use the weakest field likely in orbit (near the geomagnetic equator, $B_{\min} \approx 25 \mu\text{T}$ at 500 km):

$$m_{\text{req}} = \frac{2 \tau_{\text{dist}}}{B_{\min}} = \frac{2 \times 4.69 \times 10^{-7}}{2.5 \times 10^{-5}} \approx 0.038 \text{ A m}^2. \quad (8)$$

That's the number every axis needs to beat.

Sizing the actual hardware

X and Y use a 4 mm diameter steel core with a 14 mm outer diameter copper winding, 60 mm rod length, wound over 50 mm of it. Z runs along the 340.5 mm long axis of the 3U stack, so a rod there would be impractically long and would collide with the wheel array, avionics stack, and payload bay – an air-core coil bonded flat to the top bulkhead avoids that.

X/Y rods

Core cross-section: $A_{\text{core}} = \pi(2\text{mm})^2 = 1.257 \times 10^{-5} \text{ m}^2$. With $L = 60 \text{ mm}$, $d = 4 \text{ mm}$, the aspect ratio is $\ell = 15$, so self-demagnetization matters:

$$N_d = \frac{\ln(30) - 1}{15^2} = 0.0107, \quad \mu_{\text{rod}} = \frac{1000}{1 + 0.0107(999)} \approx 86,$$

using $\mu_r = 1000$ for the core steel. That's how a bulk permeability of 1000 collapses to an effective 86 once the rod's own shape is accounted for.

Design point: 600 turns of AWG28 magnet wire (0.36 mm insulated), 100 mA. Over the 50 mm winding length, AWG28 fits about 138 turns per layer, so 600 turns needs 5 layers – a 1.8 mm radial build-up, well inside the 5 mm annular space between core and coil OD. Resistance comes out to about 2.3Ω , so each rod draws roughly 23 mW at 100 mA, and delivers

$$m = \mu_{\text{rod}} N I A_{\text{core}} \approx 0.065 \text{ A m}^2,$$

about 70% above the 0.038 A m^2 requirement.

Z air coil

No core, so $\mu = 1$ in Eq. (2). Using an 80 mm outer diameter coil on the 100×100 mm top bulkhead, 100 turns of AWG28 at 100 mA:

$$A_{\text{coil}} = \pi(0.04 \text{ m})^2 = 5.03 \times 10^{-3} \text{ m}^2, \quad m = NIA_{\text{coil}} \approx 0.050 \text{ A m}^2,$$

about 32% above requirement. Wound as roughly 10 turns radially by 10 layers stacked (a compact $\sim 3.6 \text{ mm} \times 3.6 \text{ mm}$ cross-section), it needs about 24 m of wire, giving $R \approx 4.9 \Omega$ and $P \approx 49 \text{ mW}$ at 100 mA.

Table 2: Magnetorquer design summary.

Axis	Construction	Turns	R (Ω)	P @100mA	m delivered
X	4mm core / 14mm coil, $L=60\text{mm}$	600	2.3	23 mW	0.065 A m^2
Y	4mm core / 14mm coil, $L=60\text{mm}$	600	2.3	23 mW	0.065 A m^2
Z	80mm air coil, top bulkhead	100	4.9	49 mW	0.050 A m^2

All three clear the 0.038 A m^2 requirement with margin (Figure 3), and total simultaneous power draw is under 100 mW – a rounding error against a typical 3U power budget. In practice the axes are usually pulsed one at a time, so real draw is lower.

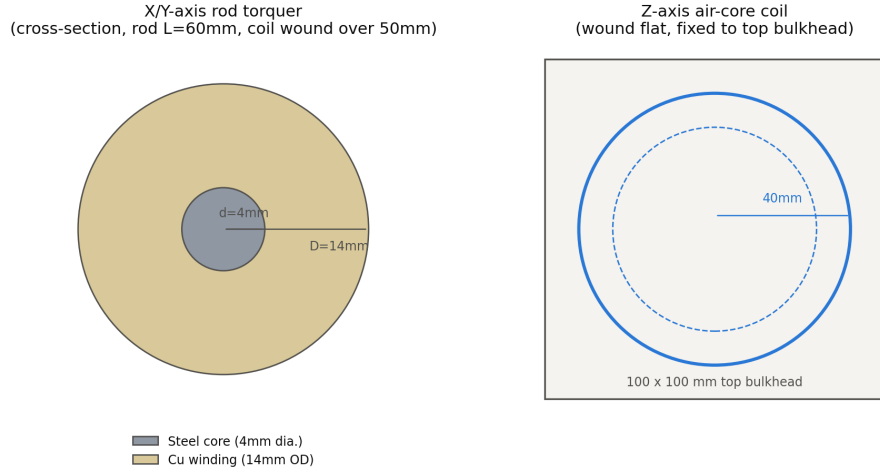


Figure 2: Left: rod cross-section. Right: the Z-axis coil's footprint on the top bulkhead.

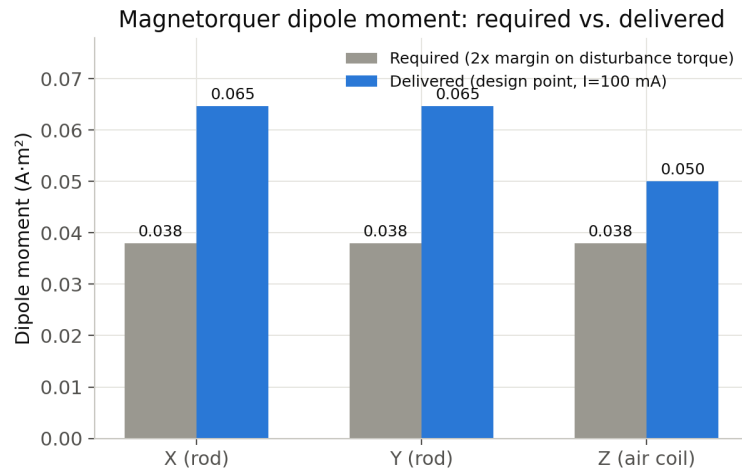


Figure 3: Required versus delivered dipole moment, by axis.